

# Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: MRBENCH | Context: Metropolitan India — Grades 1–8 Mathematics Tutoring (Student Population)

Definition: Whether the expected output modality matches regional deployment needs and accessibility.

## Justification

Output form is a LOWER-priority dimension. MRBench's evaluation outputs are categorical labels per dimension and aggregate DAMR/AC scores [Q64, Q66], not free-form text — providing actionable per-dimension diagnostics (e.g., GPT-4 reveals answer 47% of the time [Q69]) that map cleanly to deployment QA tasks. The deployment's actual product output is free-form English tutor text, which the benchmark does not directly score for free-form quality but indirectly assesses via the categorical taxonomy. Domain-agnostic NLG metrics (BLEU, BERTScore) are explicitly rejected as inappropriate [Q4, Q7, Q8, Q10] — a defensible methodological choice. The chief weakness is that DAMR does not operationalize student satisfaction or engagement, and the LLM-critic alternative is unreliable [Q107, Q109, Q114] — confirmed by 2024–2025 literature [WEB-19, WEB-20] — meaning automated re-evaluation of the deployed system using MRBench-style critics is not a valid substitute for human judgment. Partial mitigation comes from the deployment's human tutor co-authorship layer.

| ID     | Evidence                                                                    |
|--------|-----------------------------------------------------------------------------|
| WEB-19 | 2025 MDPI: LLM evaluator inconsistency confirmed for educational evaluation |
| WEB-20 | 2025 study: LLM and human evaluator scores share little common signal       |

## Strengths

- DAMR provides interpretable per-dimension percentages directly usable for model comparison and deployment QA [Q64, Q67]
- Per-dataset (Bridge vs. MathDial) reporting [Q149, Q150] enables grade-band-specific diagnostics roughly aligned with primary vs. upper-primary
- Explicit, well-reasoned rejection of NLG metrics that would be inappropriate for tutoring evaluation [Q4, Q7, Q8, Q10]
- Honest reporting of LLM-critic unreliability [Q107, Q109, Q114] prevents users from over-relying on automated re-evaluation

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                    |
|------|--------------------------------------------------------|
| OF-1 | Does output modality match regional deployment needs?  |
| OF-2 | Assess text-to-speech availability for speech output   |
| OF-3 | Consider literacy rates and accessibility requirements |
| OF-4 | Document form mismatches harming external validity     |

### Checklist responses:

OF-1: Output modality (categorical label scoring + aggregate metrics) is not the deployment's product output (free-form tutor text), but it serves as an evaluation signal that maps onto the deployment's per-dimension QA needs. The match is indirect but adequate as evaluation signal.

OF-2: Not applicable — neither benchmark nor deployment requires text-to-speech. INSUFFICIENT DOCUMENTATION on whether deployment plans audio extension; assumed text-only per regional context.

OF-3: Target population is described as fluent English speakers in metropolitan India [regional context]. Mobile internet penetration is high [WEB-10, WEB-12], so accessibility constraints around text output are minimal. INSUFFICIENT DOCUMENTATION on accessibility for students with disabilities in the deployment's target population.

| ID     | Evidence                                                             |
|--------|----------------------------------------------------------------------|
| WEB-10 | Mobile internet broadly available in Indian metros for text delivery |
| WEB-12 | ~65% of Indian children aged 5–14 used mobile phones in 2023         |

OF-4: Documented form mismatch: DAMR does not operationalize student satisfaction or engagement, which is the deployment's primary success criterion. Automated LLM critics are unreliable [Q107, Q114] and cannot substitute for human judgment in production. External validity is partially harmed: benchmark scores will not transparently predict deployment satisfaction outcomes.

| ID     | Evidence                                                                    |
|--------|-----------------------------------------------------------------------------|
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### Information Gaps

- No documented mapping between DAMR scores and student satisfaction outcomes
- No published deployment study using MRBench scores as a production QA signal

### Requires Expert Verification

- Whether the deployment's human tutor co-authors can effectively use DAMR-style per-dimension reports for QA review
- Whether deployment satisfaction signals can be calibrated to MRBench dimensions via instrumentation

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### Remediation

**Gap 1: DAMR does not operationalize student satisfaction; LLM critics are unreliable [Q107, Q114] and cannot substitute for human judgment, confirmed by 2024–2025 literature [WEB-19, WEB-20].**

Recommendation: Treat MRBench scores as one input among several to a deployment QA dashboard that also incorporates human tutor co-author ratings and student engagement signals, rather than as a sufficient indicator of deployment readiness; do not use Prometheus2/Llama-3.1-8B critics for production re-evaluation.